Does explicit news decay improve forecasts?

FININ-inspired public CMIN study | 14 October 2026 | Exploratory temporal validation

No stable, practically meaningful forecasting improvement from 48-hour decay was established.

FININ links news interactions and market-conditioned weighting to information diffusion, but does not impose an explicit age-decay law. We tested a fixed recency prior as an independent extension, keeping the next-session prediction target.

Matched experiment

Same 12-company cohort, five price sessions, eligible headlines, frozen RoBERTa and sentiment, both attention stages, and 96-value final MLP input. Equal temporal weighting gives equal mass to nonempty news buckets; decay multiplies within-bucket weights by $2^{-age/48}$ and normalizes across retained stories. Eight fits: two arms, two seeds, two chronological periods; no half-life search.

Forecast results (lower log loss is better)

Model / reference	Log loss	Accuracy
Equal temporal weighting	0.690210	54.27%
48-hour decay	0.689887	54.64%
Training-prevalence probabilities	0.689820	54.39%

Decay did not beat the training-prevalence probability reference on log loss. Both decay models in Jan-Apr 2021 predicted UP on all 984 cases; their probabilities still varied. Validation selected neural forecasts in all eight fits, with no constant fallback.

Decay minus equal-weighting log loss: -0.000324; conditional 95% interval [-0.001176, +0.000539]. The predeclared practical improvement criterion was -0.005.

Repeatability across period and seed

Evaluation period	Seed	Decay - equal loss
May-Aug 2020	17	+0.001125
May-Aug 2020	42	-0.000779
Jan-Apr 2021	17	-0.000780
Jan-Apr 2021	42	-0.000898

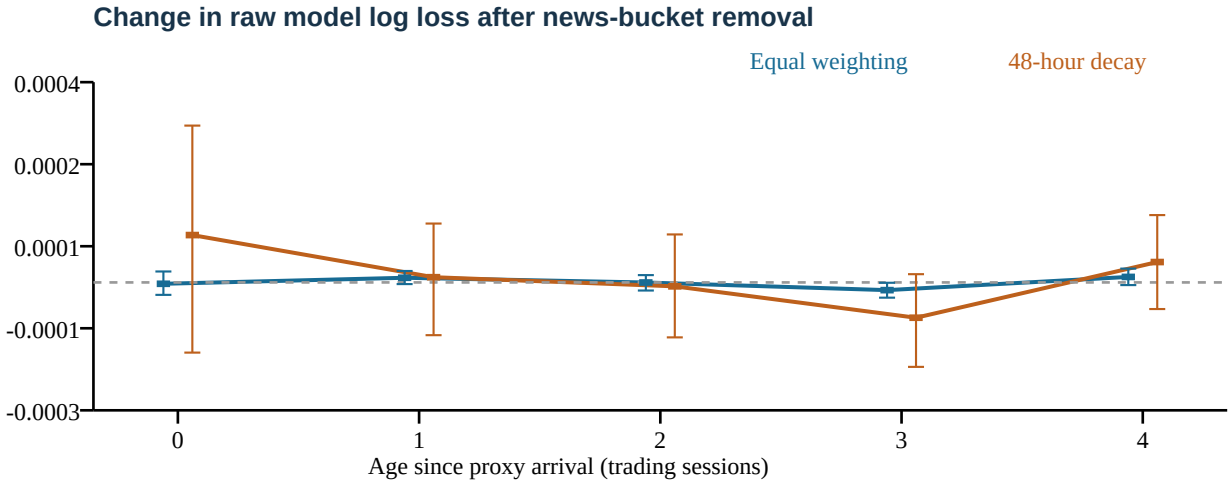
2,004 matched company-days on 167 dates. Losses average raw seed-model losses, not ensemble probabilities. Date-block intervals keep stocks together and condition on saved fits; they omit training uncertainty.

Limits on the conclusion

Existing evaluation periods were already inspected, so these are exploratory results. Source-date timing and D+2 availability are proxies. Normalized decay changes relative allocation; it does not make all old news fade absolutely. Improvement could reflect regularization or noise filtering rather than market diffusion. The existing price-only model has different geometry and is a contextual reference.

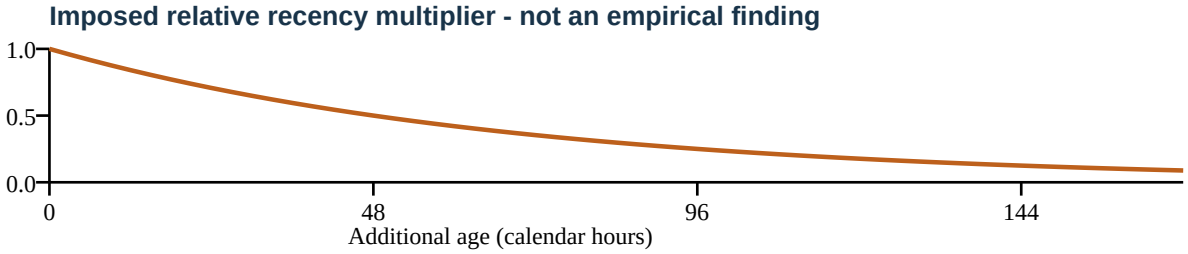
Measured news-age sensitivity

For each selected raw checkpoint, remove one arrival-age bucket, retain every price feature, and recompute predictions without retraining.



Positive values: removing that bucket worsened forecast loss. Negative values: removal improved loss. Bars are pointwise conditional 95% date-block intervals. Removal also renormalizes other news and changes coverage; these are fixed-model sensitivities, not independent age contributions or a market diffusion curve. Effects are tiny and non-monotone; nine of ten pointwise intervals include zero. The isolated equal-weighting age-3 signal does not establish a decay law.

Arrival age (sessions)	Cases with removed news	Headlines removed
0	1,344 / 2,004	5,578
1	1,341 / 2,004	5,562
2	1,334 / 2,004	5,518
3	1,339 / 2,004	5,544
4	1,340 / 2,004	5,547



The 48-hour curve was chosen before fitting. Calendar-hour decay and trading-session diagnostic ages use different clocks. No economic half-life was estimated, and this curve was not fitted to outcomes.

Connection to FININ and evidence

Wang, Cohen and Ma (2024), FININ: sections 4.2-4.3 (Eqs. 6-7), section 6.4 (diffusion interpretation). Our study tests an explicit relative-recency extension; it neither reproduces nor refutes FININ's economic claims.

Frozen protocol, configurations, all eight checkpoints, dated predictions, age-mask logits, coverage and analysis hashes: artifacts/cmin_decay_48h_2026-10-14_v1/. Negative outcomes are retained; no follow-up tuning was performed.